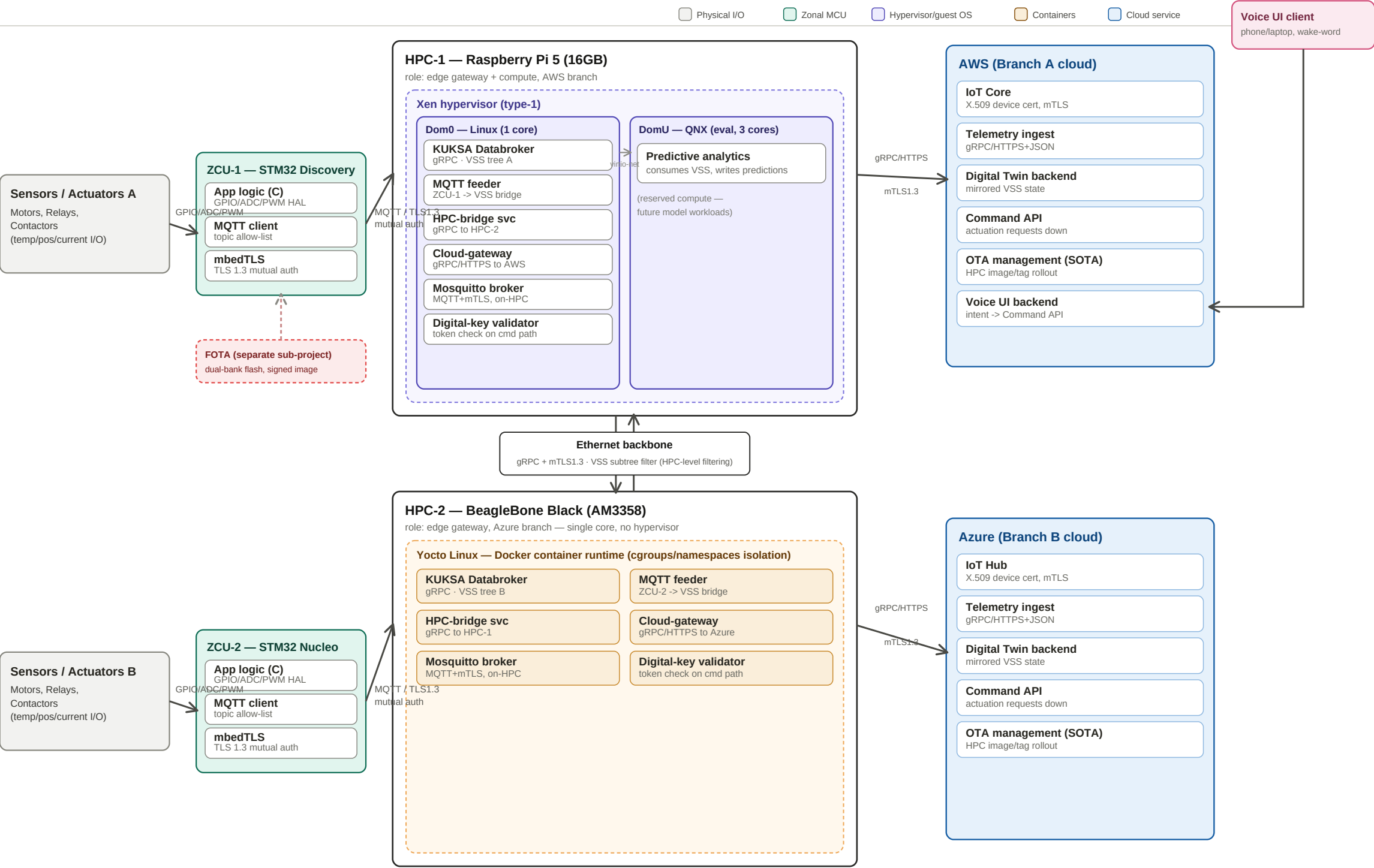


SDV_Replica_POC — System Architecture (HW + SW modules, interfaces, data flow)

Physical topology · protocol stack · security boundary — A3 landscape reference sheet



PKI / mTLS 1.3 (cross-cutting)

Root CA -> zcu-ca / hpc-ca / cloud-ca -> per-device leaf certs.
Every link above (ZCU-HPC, HPC-HPC, HPC-Cloud) is mutually authenticated against this

Filtering — two layers

Zonal level: ZCU MQTT topic allow-list (compile-time).
HPC level: KUKSA VSS subtree scoping on the HPC-bridge and cloud-gateway services.

SOTA vs FOTA

SOTA: HPC container image / rootfs update (Mender/swupdate, tag rollout).
FOTA: ZCU firmware update — separate MCU bootloader sub-project.

Digital Key

Cloud issues a short-lived signed token.
HPC's Digital-key validator checks it before a high-privilege actuation command is allowed th